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See, Touch, and Feel: Enhancing Young Children's Empathy Learning Through a Tablet Game

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ABSTRACT— Ongoing research is providing new insights into the biological rudiments of empathy and its neurobiological underpinnings. There is also growing awareness that tablet technology, when used educationally and ethically, can aid adolescents and young-adults' empathic learning. However, there has been little attempt globally to translate this new knowledge into the learning and teaching of empathy in early years education. This small-scale study aimed at enhancing 3–6-year-olds' empathy by designing a tablet game and evaluating its developmental impact by combining teachers' observation with pre-electroencephalogram (EEG) and post-EEG. Children in one Australian preschool, were invited to (1) attend to and perceive emotionally salient events in a story, (2) actively share the emotions of the characters identified, and (3) take others' perspectives, reasoning why a given emotion arises within the context. Repeated measures analysis of both EEG and observation data indicate that interacting with the tablet game enhanced participating preschoolers' empathic learning.

There is widespread recognition by education practitioners (Garnett, 2018) and researchers (Eisenberg & Morris, 2001; Hoffman, 2000; Malti, Chaparro, Zuffianò, & Colasante, 2016) that empathy should be promoted in early childhood and primary school classrooms (Berliner & Masterson, 2015). Empathy is recognized as a precursor for prosocial behaviors, such as caring and cooperation (Decety, Bartal, Uzefovsky, & Knafo-Noam, 2016; Findlay, Girardi, & Coplan, 2006), while also acting as a preventive factor against aggression (Eisenberg, Eggum, & Di Giunta, 2010) and bullying (Şahin, 2012). It is also claimed that academic success requires an empathic values dimension (Duncan & Sankey, 2019; Toomey, Lovat, Clement, & Dally, 2010), not least in promoting effective classroom collaboration and communication, which can be essential for learning. Given the educational and social importance of empathy, it is clearly paramount that early childhood and primary school educators begin sowing the seeds of empathy in young minds. In terms of educational practice, the key issue is how best that might be achieved.

Findings from developmental research have shown that, in the early years, there is rapid development of components involved in empathy, including (1) affective sharing, (2) emotion understanding, and (3) emotional regulation processes (Decety, 2010; Decety & Jackson, 2004). During the ages of toddlerhood and preschool, there is a remarkably rapid development of empathy, as the three components noted above mature. Between the ages of 2 and 3 years, as emotion understanding develops rapidly (Decety, 2010), young children become better at apprehending others' inner states using varied cues, making distinction between others' emotion state and their own (Vaish & Warneken, 2012). Also, as top-down regulative processes start to mature, children's prosocial reactions become selective. For example, 3-year-old children take another person's intention into account when deciding whether to help the person or not (Vaish, Carpenter, & Tomasello, 2009).

From the perspective of complexity science, empathy emerges from the utter complexity of the recursively dynamic, self-organizing brain (Kim & Sankey, 2009, 2010), when, for example, bottom-up process (e.g., affective arousal, emotion awareness, and understanding) and

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top-down process (e.g., emotion regulation, recognizing perceiver's motivation, memories, intentions, and attitudes) are combined, yielding insights into the thoughts, and feelings of others (Decety, 2010; Decety & Jackson, 2004). Studies in developmental neuroscience have shown how partially distinct, yet recursively interacting, neural mechanisms mediate empathy development in early years (Decety, 2010; Decety & Michalska, 2012). Although the number of neuroimaging studies concerned with young children's empathy is limited, no doubt partly as result of methodological constraints, a few seminal studies involving non-invasive and child-friendly methodologies (e.g., functional near-infrared spectroscopy [fNIRS], electroencephalogram [EEG]) are shedding light on the neural mechanisms underlying children's perception of other people's distress. A fNIRS study by Brink et al. (2011) found increased activation of the medial orbitofrontal cortex (OFC), between the age of 4 and 8 years, when children engaged with tasks involving affective empathy. Decety and Michalska's (2010) cross-sectional fMRI study reaffirms the recruitment of OFC when perceiving intentionally inflicted harm. However, when investigating across an age range of 7–40 years, they found activation in the OFC shifted from its medial portion in young participants to a lateral portion in older participants, reflecting a gradual shift from monitoring of visceral and motor responses to a more evaluative processing of emotion.

Several event-related potential (ERP) studies of perception of other's emotion and distress have shown that empathic dispositions/behaviors are related to neural dynamics indexed by components such as P200 and late-positive potential (LPP), but not early automatic components (e.g., early posterior negativity [EPN], N2) (Chiu Loke, Evans, & Lee, 2011; Cowell & Decety, 2015b; Decety, Meidenbauer, & Cowell, 2018). P200 is an attention-related component and sensitive to emotionally salient visual stimuli. P200 amplitudes at frontal and central sites are higher with negative stimuli than with positive or neutral stimuli, showing the "negativity bias" (Carretié, Martín-Loeches, Hinojosa, & Mercado, 2001; Carretié, Mercado, Tapia, & Hinojosa, 2001). In typically developing preschool aged children, P200 amplitudes are greater for "antisocial" than for "prosocial" actions (Meidenbauer, Cowell, & Decety, 2018). In adult population, conditions such as phobic (Sarlo & Munafò, 2010), high-level of callous-unemotional (CU) traits (Brislin et al., 2018) and "emotional desensitization" resulting from exposure to violence (Stockdale, Morrison, Kmiecik, Garbarino, & Silton, 2015) are known to lead to reduced P200 in response to distress cues (e.g., expression of fear), indicating that sustained attention and processing is precluded. LPPs are known to index top-down cognitive reappraisal and regulation of emotion arousal (Dennis & Hajcak, 2009; Hajcak & Dennis, 2009; Hajcak, Weinberg, MacNamara, & Foti, 2012). Cheng, Chen, and Decety's (2014) study, with children aged 3–9 years old, has shown that the difference wave of N2 between pain and neutral stimuli, indexing empathic arousal, has an age-related decrease while the difference wave of LPP has an age-related gain. Also, difference amplitudes of LPP (help-harm) predicted sharing behavior in preschool aged (3–5 years) children (Cowell & Decety, 2015b).

Findings from developmental sciences, including neural mechanisms that underpin empathy development and the normative trajectory of changes in neurotemporal dynamics, provide helpful pointers that educators can deliberate on when both designing pedagogical approaches to enhance empathy and assessing the outcomes of pedagogical implementation. However, there is growing concern that existing empathy intervention programs are not systematically translated from developmental research (Malti et al., 2016). A research-informed empathy education that caters for relative and differentiated needs in different developmental age-phases would likely to be more effective. If there is a discrepancy between a child's actual empathic capacity and educator's expectation of such capacity, it may compromise the effectiveness of the intervention (Noam & Hermann, 2002).

Against these various research backdrops and in response to the practical issue of how educators might begin sowing the seeds of empathy in young minds, the present study aimed at enhancing empathy and active perception of other's states in a small cohort of 3–6-year-olds, by designing a tablet game and evaluating its developmental impact. Many studies of emotion stress the igniting-role played by "perceiving," that is, discerning critical signs and cues associated with emotional instances, in affective processes generally (Barrett, 2017; Zonneveld, Platje, Sonneville, Goozen, & Swaab, 2017), and specifically in empathically motivated prosocial behavior (de Waal & Preston, 2017; Decety et al., 2016; Preston & de Waal, 2002, 2017). Ontogenically, early years are a very important period when children learn how to regulate their visual attention to interpret important social and emotional cues, while avoiding less important or distracting visual cues in complex social situations (Poulin-Dubois et al., 2018; Sankey & Kim, 2016; Teufel, Fletcher, & Davis, 2010). Tablet games hold great potential promise in fostering young children's empathic perception skills; offering young children opportunities to engage with authentic problems that are common in a child's life, while attending to various real facial and emotional expressions (e.g., face of sad boy) and contextual information (e.g., one child's monopoly of toys).

There is a growing literature reporting that purposefully designed prosocial games can enhance levels of empathy in adolescents and young adults (Greitemeyer, Osswald, & Brauer, 2010), by altering empathy-related circuits in the brain (Kral et al., 2018). However, to our knowledge, there is

no empirical evidence of how purposefully designed tablet games might impact empathy development at the preschool age. The present study developed a tablet game that invited preschool children to (1) attend to and perceive emotionally salient events in a story, (2) actively share the emotions of the characters identified, and (3) take others' perspectives, reasoning why a given emotion arises within the context. The impact of empathic learning was investigated at two levels, by combining teacher's observation of empathic engagement displayed by children and EEG/ERPs responses from children. Based on aforementioned evidence on ERP correlates of empathy development, we hypothesized that children's enhanced empathic perception and extended attention, resulting from the game engagement, will be indexed by increased amplitudes of P200 and LPP in response to antisocial situations. Additionally, we hypothesized that improved empathic perception will be overtly expressed into positive behaviors observable by the teachers.

METHOD

Participants

Twenty-six Australian born preschool-age children from one preschool located in Sydney, Australia, participated in the study. Participating children were between the ages of 43 and 62 months ($M = 51.51$ months, $SD = 5.51$; $N = 8$ girls). Data were collected 3 days per week, over 10 weeks using two classrooms of children that were in their normal mixed ability, parallel age groups. One served as an intervention group ($N = 12$, 3 girls); the other as a control group ($N = 14$, 5 girls). This classroom-based condition assignment enabled the researchers to run "blinded" control within the same preschool, within the approved ethics requirements of the Human Research Ethics Committee (HREC). Out of the total 26 children, 22 had both pretest and posttest EEG data, whereas four children's EEG data were not completed, because they withdrew from the pretest or posttest ($N = 3$), sick leave during posttest ($N = 1$), not completing posttest ($N = 1$).

Tablet Game Intervention

To enhance children's empathic learning, the research team developed *The Empathy World*, a story-based tablet game (see Figure 1). During the 10-week intervention period, children in the experiment group engaged with a total of 30 episodes under the four contextual themes (Nursery, Kindergarten, School, and Home) that are especially familiar to young children. The familiarity of environments aimed at proficient transfer of the empathic learning gained from the game to real life. Starting from the most familiar nursery episodes in Week 1, children progressed to episodes with increased complexity (e.g., engaging with older children and adult characters in more complex social situations). Regarding emotional tones and valences, the game was balanced by including episodes of both prosocial and antisocial interpersonal interactions. Also, the children engaged with both basic (e.g., anger, sadness, happiness, fear) and complex emotions (e.g., loneliness, gratitude, pride, frustration) with the latter occurring in later weeks when children engaged with more complex stories.

In each episode, the initial scene invited young children to "notice" interpersonal cues (e.g., crying baby in Image 2, Figure 1). If a child gives primacy to noninterpersonal cues (e.g., touching the colorful toys in the sandpit while ignoring the crying face, Image 2), the child was given a reinvitation to go back to the scene. Once the child had perceived the socially salient cues, the following scene invited the child to guess the feelings of the character in the given circumstance (Image 3, Figure 1). When a child chose the most important (appropriate) item on the scene, a positive audio-visual feedback was given.

For the first 5 weeks, when children engaged with Nursery and Kindergarten themes, each day's game play was followed by a verbal discussion with a researcher. This discussion encouraged children's perspective-taking by inviting them to tell the researcher why they thought the character felt certain emotions in the given context. In Weeks 6–10, when school and home themes ran, a guided perspective-taking opportunity was built in the game; hence, the last scene of each episode (e.g., Image 6, Figure 1) invited the child to give reasons why the character in the previous scene felt the emotion she/he chose.

The audio-visual feedback in the game was designed to give not only attentional guidance and positive reinforcement, but also to help children discern the subtle differences between emotions and contexts. For example, in the episodes that children encountered in Weeks 6–10, the guiding feedback for inappropriate responses encouraged the children reevaluate the feelings making a comparison with what they had learnt before. For example:

> We can feel ____ (emotion chosen by a child) in these stories (previous stories that children already engaged in the game), but would you feel _____ in this story (presenting the current story again to children)? Try again?

As noted, two conditions were assigned based on classroom and randomization was not implemented, as it would have disrupted the children's sense of teacher/group membership. Adopting a classroom-based assignment enabled us to "blind" the children in the control group to their condition by physically dividing two groups in separate classroom venues. It was also considered ethically important, as it minimized the negative experience (e.g., frustration, jealousy,



Fig. 1. Images captured from the game. *Note.* All educational elements are clickable and accompanied with audio guidance, recorded by a fully qualified female teacher. Images: (1) Theme map with venues of different episodes; (2) an example of perception-action selection task from the Nursery theme. The "glow effect" appears when a child touches an item. If an item is clicked, audio-visual feedback is given; (3) emotion-sharing component for learning a basic emotion (sadness) where the cartoon face options are given; (4) perception-action selection task from the home theme; (5) Emotion-sharing component for learning a complex emotion (thankfulness) where real human face options are given; and (6) perspective-taking component where children are invited to reason why the character felt thankful.

confusion) children might experience from unequal treatment and disruptions within the preschool resulting from other types of condition assignment. A major consideration in designing the research was the ethical issues that arise when moving laboratory-based technologies into real-word classroom environments. As Uncapher (2019) has recently noted, the researcher's "assessments and/or interventions may cause harm or raise any ethical concerns when conducted in school" (p. 64). For equity reason, the HREC further required that the control group was given opportunity to engage with the game upon the conclusion of the data collection. The researchers implemented this protocol.

In the experiment group, to standardize the quality of the game-learning process, each child played the tablet game on an individual basis, with the assistance of a researcher. Children did not take the tablet from the preschool to play the game at home. This individualized learning with adult researcher support aimed to ensure all children in the experiment group fully and evenly engage with every aspect of the game learning, regardless of their tablet skills and attention span. Within the preschool, a dedicated space equipped with a table, child chair, and video camera was set up. The researcher invited each child to this game-learning venue when it was his/her turn. Cooperation and group learning was not included in the game implementation. The researcher ensured that each child fully engaged with the game for a minimum 10 minutes each time.

Questionnaires

Pretest and posttest observation of children's empathic behavior was gathered from their teachers. As the administration of two conditions was noticeable (e.g., each child in the experimental condition had to leave the classroom activities while participating in the game-learning), teachers could not be blinded to the condition. However, given the nature of the questionnaire, teachers were considered more reliable informants than the parents, or the blinded assessor. (Unlike most parents, teachers, in their everyday practice, constantly observe children when they engage with their peers, and thus are able to compare each child's empathic capacity against a baseline of many other children.) Two teachers (one teacher from control group; one teacher from experimental group) completed one copy of the empathy questionnaire (EmQue; Rieffe, Ketelaar, & Wiefferink, 2010) for each child. The EmQue is a 20-item

observer-report questionnaire, representing three facets of empathy observable in young children: (1) emotion contagion (seven items, α = .662), (2) Attention to others' feelings (seven items, α = .814), and (3) prosocial actions (six items, α = .834). Teachers rated the degree to which each item, reflecting a type of behavior, applied to each child in her classroom over the past 2 months on a 3-point scale (0 = never, 1 = sometimes, 2 = often).

The Chicago Moral Sensitivity Task and EEG Recording

To investigate changes in perceptual sensitivities to actions that might lead to positive or negative emotions, at a temporal neurodynamics level, participating children were invited to complete the Chicago Moral Sensitivity Task (CMST) before and after the 10-week intervention. In the CMST, children watched visual scenarios of characters intentionally performing either prosocial acts (e.g., sharing food) or antisocial acts (e.g., throwing a snowball to the other character's face) (Cowell & Decety, 2015a, 2015b). To avoid the risk of producing artifacts and other confounders, children were not asked to give any response, such as pressing keys or mouse buttons that require physical movement; rather, they passively viewed the cartoon scenarios. The version of the CMST used in this study included 120 trials (60 prosocial and 60 antisocial), 3 s in duration per scenario, randomly presented, with a jittered intertrial interval of 500–1,500 ms. ERPs were time-locked to the onset of the second picture. EEG data were collected from a 32-channel active electrode system (actiCHamp, Brain Products, Germany) at 2 KHz, referenced to Cz, and all electrodes impedances were kept below 30 kΩ. Each child's attentiveness was monitored through the eye-tracking data.

Data Analysis

ERPs Data Analysis

EEG data preprocessing was performed with Brain Vision Analyzer (Version 2.1.1) software (Brain Products) following Cowell and Decety (2015a, 2015b). Data were down sampled offline (256 Hz) and referenced to the average of all electrodes. Following referencing, EEG data were filtered using an IIR filter of 1–30 Hz. Artifact rejection was carried out using a −200 μV to 200 μV threshold and visual inspection and rejection. Gratton-Coles ocular artifact correction was performed on EEG data to identify and correct ocular movements, seeded off electrode Fp1 (Gratton, Coles, & Donchin, 1983). Following basic EEG preprocessing, data were segmented according to stimulus type (harming/helping). Epochs were created with 200 ms baselines and 1,000 ms stimulus presentation following the onset of the second stimulus frame. Epochs were averaged by trial type and were baseline corrected. Corrected averages per individual were combined for a grand average per individual and ERPs were analyzed across subjects by trial types and experimental conditions. Only children who had at least 20 artifact-free trials for each condition (harming and helping) were included in the grand average and further analyses ($n = 20$). Average number of trials used in final analysis did not differ by group or time.

Existing publications (e.g., Cowell & Decety, 2015a, 2015b; Meidenbauer et al., 2018) served as a priori information to guide the choice of latency windows and electrode sites. This was possible as the recordings were performed in the same condition reported in these publications (same EEG hardware, same sampling rate, reference, impedance threshold level etc.). The following components and time windows were the focus of analysis: early posterior negativity (EPN, 100–175 ms), P2 (150–350 ms), N2 (200–400 ms), LPP (400–600 ms), later late positive potential (600–800 ms), slow waveform (SW; 800–1,000 ms). Given our interest in understanding how the game intervention influences perceptual sensitivities to actions that might lead to positive or negative emotions, we focused our analyses on the difference waves generated by subtracting harming from helping situation (i.e., $P2_{diff} = P2_{Help} - P2_{Harm}$).

In the present study, nine electrodes were included in the analyses: anterior (F3-Fz-F4), central (C3-Cz-C4), and posterior (P3-Pz-P4) sensors. Electrodes were not averaged by region. Numerous neurophysiological studies have documented that individual differences in the quality and intensity of one's response to affectively evoking stimuli are related to cerebral asymmetry in brain functions (e.g., Davidson, 1992; Wheeler, Davidson, & Tomarken, 1993). Averaging electrodes from left and right hemispheres dampen out the individual differences in affective responses to the stimuli. Therefore, we included locations of electrodes (left, central, and right) within each region as a within-subject variable. Mean values of difference amplitude (e.g., EPN_{diff}, $P2_{diff}$, and LPP_{diff}) were analyzed in each of the three regions by repeated-measure analyses of variance (ANOVA), with Group (control and experiment) as the between-subject factor, plus Time (pre and post) and Electrodes (left, central, and right) as the within-subject factors. Greenhouse–Geisser correction was used where appropriate. For ANOVAs, partial eta squared (η_p^2) was used to measure effect sizes. Post hoc comparisons or simple effect analyses were performed with a Bonferroni's correction.

RESULTS

Changes in Empathic Behavior Observed by Teachers

Mean ratings for each scale of the EmQue are provided in Table 1 and Figure 2. Initially, preliminary *t*-tests were conducted to confirm that there were no significant differences in pretest ratings. The preliminary analyses revealed

Table 1
Mean (*SD*) Values for Three Subscales of the EmQue

Subscale	*Time*	*Experiment (N = 14)*	*Control (N = 12)*
Emotion contagion	Pre	7.83 (2.03)	6.64 (.84)
	Post	7.25 (2.52)	6.36 (1.08)
Attention to others' feelings[a]	Pre	14.92 (2.39)	17.29 (3.07)
	Post	17.50 (1.67)	14.50 (2.53)
Prosocial actions[b]	Pre	10.33 (2.46)	10.71 (3.93)
	Post	12.75 (2.59)	12.14 (3.05)

[a]Time × Group interaction effect $p = .000$.
[b]Time main effect $p = .005$.

no group differences for two scales (emotion contagion and prosocial actions), but a significant difference for attention to others' feelings, $t(24) = -2.164$, $p = .041$.

A repeated measure multivariate analysis of variance revealed that there were significant time main effect, $F(3, 22) = 3.548$; $p = .031$; $\eta_p^2 = .326$, and Time × Group interaction, $F(3, 22) = 8.300$; $p = .001$, $\eta_p^2 = .531$, in the overall empathy. Univariate tests were conducted for each subscale, using Bonferroni adjusted alpha levels of .017 per test (.05/3).

Emotion Contagion
No time main effect was observed, $F(1, 24) = .889$; $p = .355$, $\eta_p^2 = .036$. No Time × Group interaction was found either, $F(1, 24) = .104$; $p = .750$, $\eta_p^2 = .004$.

Attention to Others' Feelings
A significant Time × Group interaction was found, $F(1, 24) = 22.893$; $p < .001 < .017$; $\eta_p^2 = .49$. Paired *t*-tests indicated significant increase of scores in the experiment group, $t(11) = -2.745$, $p = .019$, and significant decrease for the control group, $t(13) = 4.246$, $p < .001$. No time main effect was observed, $F(1, 24) = .033$; $p = .858$, $\eta_p^2 = .001$.

Prosocial Actions
A time main effect was observed, $F(1, 24) = 9.620$; $p = .005 < .017$, $\eta_p^2 = .286$, indicating significant increase of scores in both groups. However, no Time x Group interaction was found, $F(1, 24) = .635$; $p = .433$, $\eta_p^2 = .026$.

Changes in ERP Responses after the Game Intervention
As mentioned earlier, we focused our analyses on the difference waves generated by subtracting harming from helping situation (Figure 3). This subtraction enabled us to see the changes in preferential neural processing, or bias, associated with prosocial and antisocial actions.

EPN Component (100–175 ms)
Neither main nor interaction effect were found in three regions. This result implies that the extent of changes in relatively automatic selective attention (i.e., EPN_{diff} values) did not differ between control and experiment groups.

P2 Component (150–350 ms)
In all three regions, Time × Group interaction was significant: frontal, $F(1, 18) = 4.687$; $p = .044$, $\eta_p^2 = .207$; central, $F(1, 18) = 7.217$; $p = .015$, $\eta_p^2 = .286$; posterior, $F(1, 18) = 5.443$; $p = .031$, $\eta_p^2 = .232$. Post hoc tests, using Bonferroni adjusted alpha levels of .006 per test (.05/9), confirmed that Time × Group interaction was particularly significant at following two midline electrodes: Cz, $F(1, 18) = 13.795$; $p = .002$, $\eta_p^2 = .43$; Pz, $F(1, 18) = 11.989$; $p = .003$, $\eta_p^2 = .40$. At Fz, Time × Group interaction was

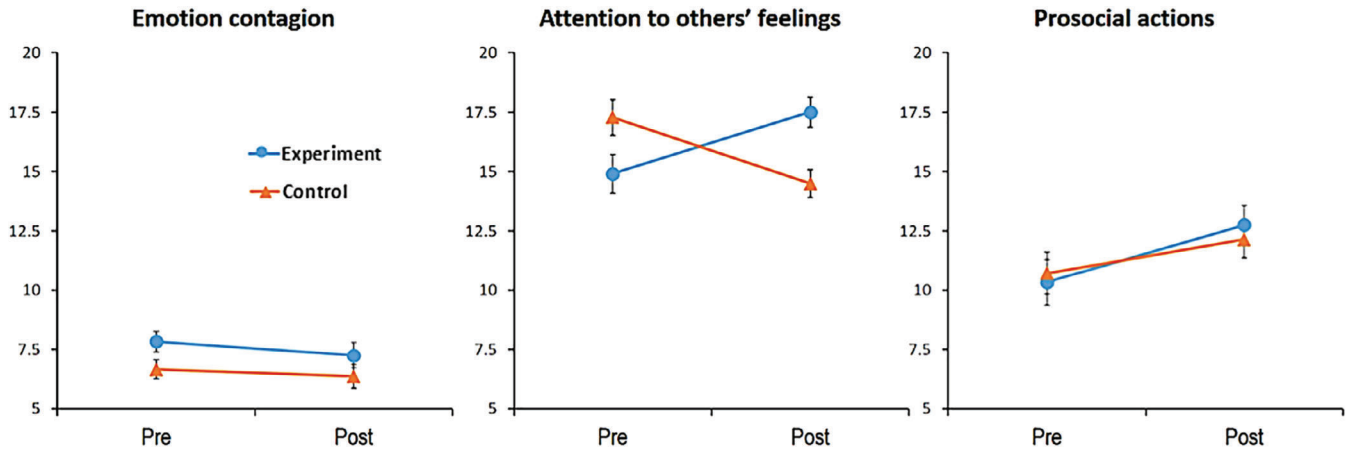


Fig. 2. Mean scores for three subscales at pretest and posttest across the game intervention and control groups. These data are scores for all participants who participated in the study ($n = 14$ for Control; $n = 12$ for experiment). Error bars indicate SEs.

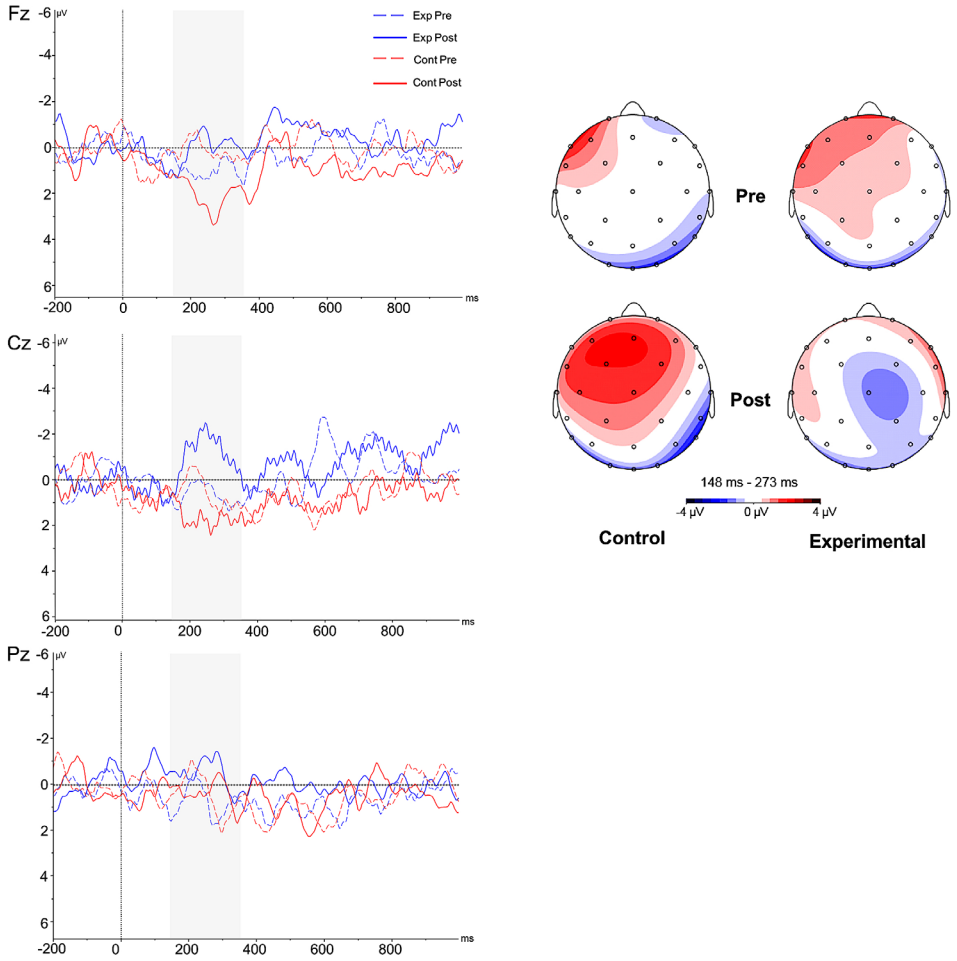


Fig. 3. P2 Difference Waveforms from three midline electrodes (Fz, Cz and Pz) (left) and range-scaled voltage spline maps of the scalp distribution of P200diff (right). Note. 1. Difference waveforms (left) were generated by subtracting amplitudes of harming situation from helping situation, with negative values plotted upwards. The colours of difference waveforms (blue vs. red) represent two group conditions (experimental vs. control), while the solidity of waveforms (dotted vs. solid) represents the timing of ERP recordings (pretest vs. posttest). 2. On the right are range-scaled voltage spline maps of the scalp distribution of P200diff in four conditions: control group at pretest (top-left); experimental group at pretest (top-right); control group at posttest (bottom-left); and experimental group at posttest (bottom-right).

significant at a lenient threshold ($p < .05$, uncorrected), $F(1, 18) = 5.089$; $p = .037$, $\eta_p^2 = .22$. Comparison of difference amplitudes from Cz revealed that difference amplitude decreased in the experimental group (from .137 μV to −1.90 μV) while it considerably increased in the control group (from .604 μV to 4.952 μV). This result suggests that, at the time of posttest, two groups tended to have different processing bias: preferential processing of "harming" actions in the experimental group; preferential processing of "helping" actions in the control group.

N2 Component (200–400 ms)

In the frontal region, Time × Group interaction, $F(1, 18) = 4.795$; $p = .042$ $\eta_p^2 = .210$, and Time × Electrodes interaction, $F(2, 17) = 4.516$; $p = .027$, $\eta_p^2 = .347$, were significant. However, post hoc comparisons did not confirm statistical significance between pre- and posttests in the experimental group ($ps > .05$). In the central region, only electrode main effect was significant, $F(2, 17) = 3.927$; $p = .040$, $\eta_p^2 = .316$. No main or interaction effect was found in the posterior region.

LPP Component (400–600 ms)

In the frontal region, the Time × Group interaction was significant, $F(1, 18) = 5.219$; $p = .035$, $\eta_p^2 = .225$. In the frontal region, while the mean LPP_{diff} remained close to zero in the control group (−.825 μV to .415 μV), the mean LPP_{diff} decreased (.384 μV to −1.068) in the experiment group, indicating preferential processing of "harming" actions. However, the statistical significance of this change was

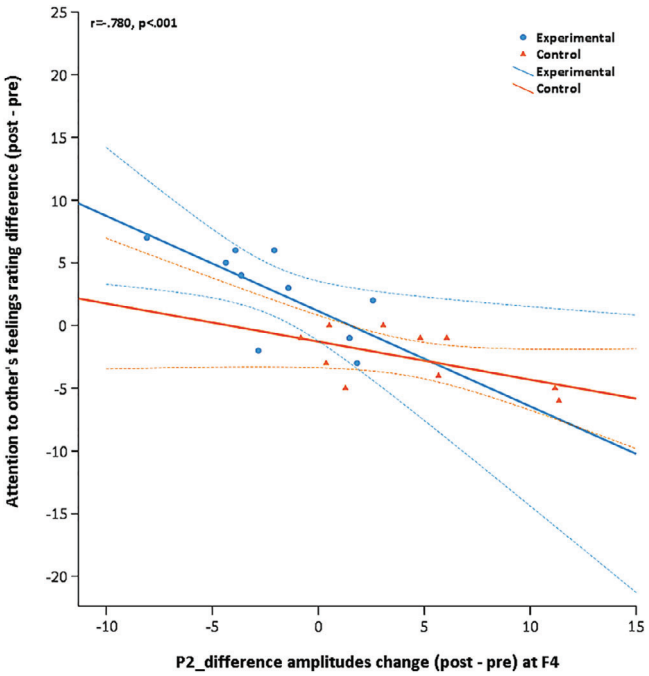


Fig. 4. Changes of $P2_{diff}$ at F4 predict changes in attention to others' feelings.

not confirmed in post hoc tests (*p*s > .05). In the posterior region, electrode main effect, $F(2, 17) = 4.270$, $p = .022$, $\eta_p^2 = .192$, was significant.

Late LPP Component (600–800 ms)
No significant main or interaction was found.

SW Component (800–1,000 ms)
The Time × Group interaction was found in the central region, $F(1, 18) = 4.636$, $p = .045$, $\eta_p^2 = .205$.

Relationship between Changes in Children's ERPs and Empathic Behavior Observed by Teachers
Changes of $P2_{diff}$ at F4 were negatively correlated with changes in teacher's ratings on attention to others' feelings ($r = -.780$, $p < .001 < .05/9$ with Bonferroni correction, see Figure 4). This negative relationship was greater in the experiment group ($\beta = -.759$, $R^2 = .484$) compared to the control group ($\beta = -.303$, $R^2 = .334$). Also, changes of SW_{diff} at Cz showed a marginal negative correlation with changes in teacher's ratings on attention to others' feelings ($r = -.498$, $p = .025 > .05/9$).

DISCUSSION

The findings have demonstrated some important developmental benefits of engaging with a tablet-based empathic learning. Teachers reported that children who engaged with the tablet game became more attentive to other's feeling, compared with those who did not. Also, this increased attentiveness showed a negative correlation with changes in P200 difference waveforms, indicating that a child whose sensitivity to interpersonal harm has been heightened at a neurotemporal level ($P2_{diff}$) has also become more attentive to other's feelings. Thus, the ERP data corroborate the observations of the teachers that learning experience of engaging with the *Empathy World* game has enhanced children's empathic perception and their attention to others' distress.

From the perspective of neurobiology, a number of commentators have stressed the importance of "perception" and "perceptual categorization" as a catalyst to learning, affective experience and development (Barrett, 2017; Edelman, 1987; Thelen & Smith, 1994). In empathy studies, it is widely recognized that perceiving other's distress often initiates empathic concern (Zahn-Waxler, Radke-Yarrow, Wagner, & Chapman, 1992) and motivates prosocial behaviors (Decety et al., 2016; Decety et al., 2018). Preston and de Waal proposed that at the core of emotional contagion is the perception-action mechanism (PAM) (de Waal & Preston, 2017; Preston & de Waal, 2002, 2017), wherein "perception selects elements in the environment that require or suggest a response" (Preston & de Waal, 2002, p. 6). Moreover, developmentally, a child's ability to recognize, discriminate, and understand others' emotion occurs early, with some reaching adult levels by approximately 4–5 years of age (e.g., Nelson, 1987). This suggests that pedagogical intervention aimed at encouraging empathic perception should occur in preschool age when children extensively learn how to discern important cues from complex social world (Poulin-Dubois et al., 2018; Teufel et al., 2010). The older a child becomes, the greater the cost might be because new empathic learning would require "de-categorization" or "recategorization" of prelearned perceptual categorization.

In the design of the game, we purposefully built in tasks that employ the PAM, where children are encouraged to select (e.g., click and touch) items that are more likely to require a response in real life situations (e.g., crying baby) rather than other attractive items that compete for their attention (e.g., colorful toys). Also, the choice of the CMST, for the ERP data collection, helped us to capture the enhanced perceptual acuity at the neurotemporal level. It should be noted that while many experimental tasks concerned with empathy tend to use decontextualized facial images (with extremely salient emotions) or highly overt images of potential or actual physical pain, the CMST is contextualized and invites child's perceptual categorization of helping and harming actions from subtle cues in semi-real cartoonized images.

In line with our hypothesis, P2 served as an index of perceptual sensitivity to harming actions, which is consistent with findings from Meidenbauer et al. (2018). In nonaffective ERP studies, P2 is more positive when the target stimuli are infrequent (Luck & Hillyard, 1994; Mangun & Hillyard, 1995). In affective visual ERP studies, P2 amplitude at frontal and central sites indexes "negativity bias" (Carretié, Martín-Loeches, et al., 2001; Carretié, Mercado, et al., 2001), which operates at the phase of automatic evaluative-categorization. The developmental trajectory of P2 is not well known. However, in preschool-aged children, P2 amplitudes are greater when viewing "antisocial" actions than "prosocial" actions (Meidenbauer et al., 2018). In adult populations, reduced P2 to negative stimuli (e.g., fearful or distressed faces) are associated with CU traits (Brislin et al., 2018) or "emotional desensitization" (Stockdale et al., 2015). Taken together, findings from this study imply that, after engaging with the empathy game, children's attentional resources are more easily mobilized when encountering antisocial or harming actions. Research on relationship between empathy and prosocial behavior highlights that witnessing other's distress often leads to preclusion of sustained processing in frontal cortex (Decety et al., 2016). At a neurobiological level, this preclusion seems to be indexed by reduced P2 (Sarlo & Munafò, 2010), while at an overt behavioral level, aversion to the situation or increased self-concern is expressed (Kim & Chang, 2014; Stephan & Finlay, 1999). In young children, in addition to gaining perceptual acuity to the other's needs, the developmental task is to gain resilience to stress and fear that arises while witnessing others in need (Decety et al., 2016). If their attentional resources are mobilized and not precluded, perceived distress is more likely to lead to empathic concern or motivation directed at improving the well-being of the other (Decety et al., 2016).

This study provides provisional but also promising evidence that tablet games can be a very productive resource in preschool classrooms, in aiding young children's empathic learning; especially their perceptual acuity to the other's needs, which is a precursor to empathic concern (Decety & Cowell, 2014). While there is call for empathy pedagogies that are systematically translated from developmental research (Malti et al., 2016), preschool teachers often have difficulty finding resource and time to design their own materials (Berliner & Masterson, 2015). This study suggests that tablet-based empathy learning games can help meet this need, especially if the pedagogical design of the game is collaboration between teachers and researchers, as it was in designing the *Empathy World*. Furthermore, as in the case of this study, carefully designed games can serve as research tools, allowing researchers to investigate developmental trajectories of different components in empathy and their relationship with children's social engagement in real classrooms.

We are fully aware that the small sample size, clustered nature of the data, baseline difference between two groups in the present study put limits to our capacity to extrapolate the findings from statistical results. Nevertheless, the experience of conducting the research described in this article has convinced the authors of the merits of this research focus for future studies. A confirmatory study with larger numbers of participants is now warranted. Also, any follow-up study would need to investigate the necessary intensity and duration of game-intervention to reach the changes in children's real-life empathic engagement.

Acknowledgment—This work was supported in part by intramural grants from the China Studies Centre and Faculty of Arts and Social Sciences, the University of Sydney. Both authors contributed to the conception, design, data collection, analysis, and writing of the manuscript.

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